Fall 2026 Semester

Cryptographic Application

KOREA UNIVERSITY

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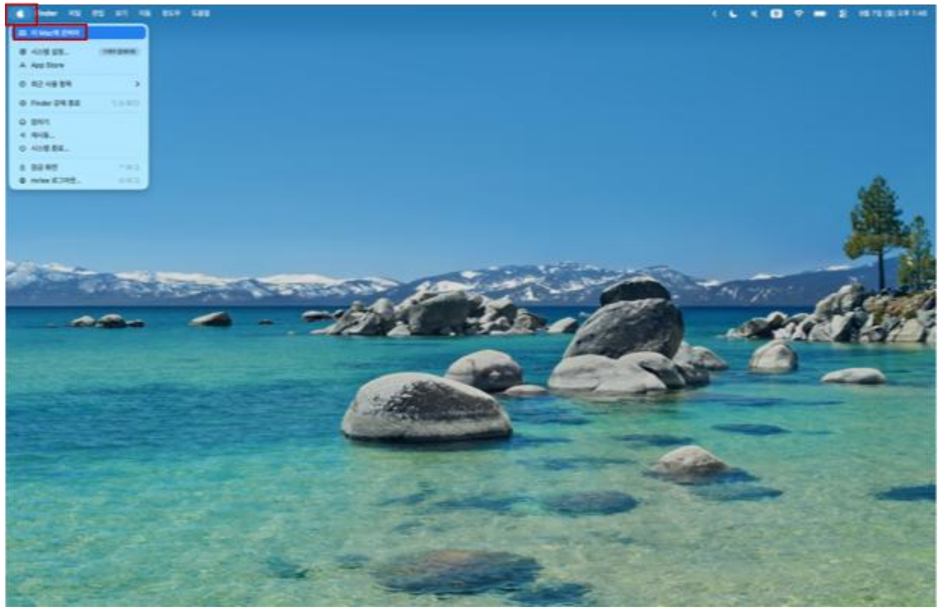
Setting for MacOS

Wonsuk Choi
beb0396@korea.ac.kr
School of Cybersecurity
Korea University, Rep. Korea

1. Check the Mac Processor

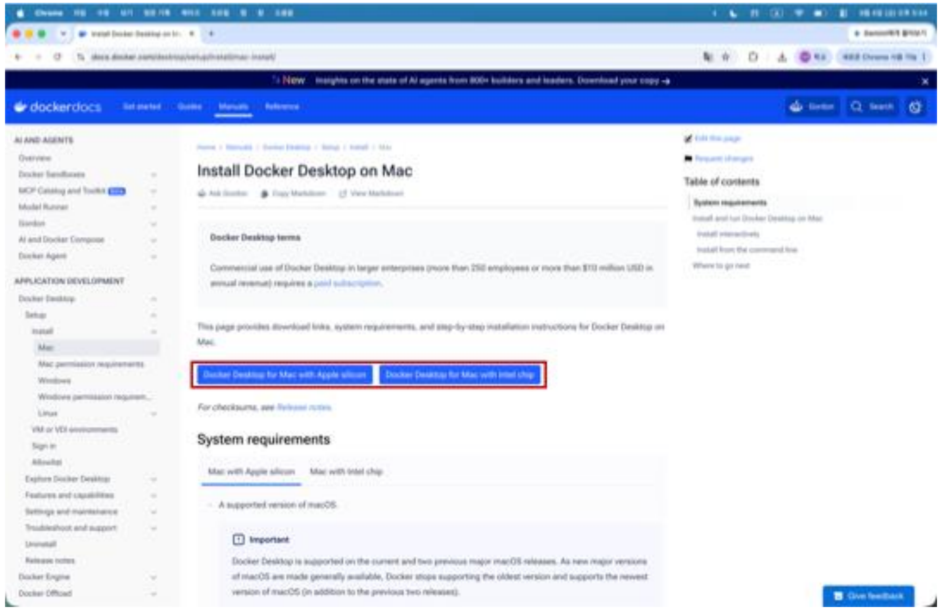
- Select Apple menu > About This Mac.
- Check whether the Mac uses an Apple chip or an Intel processor.

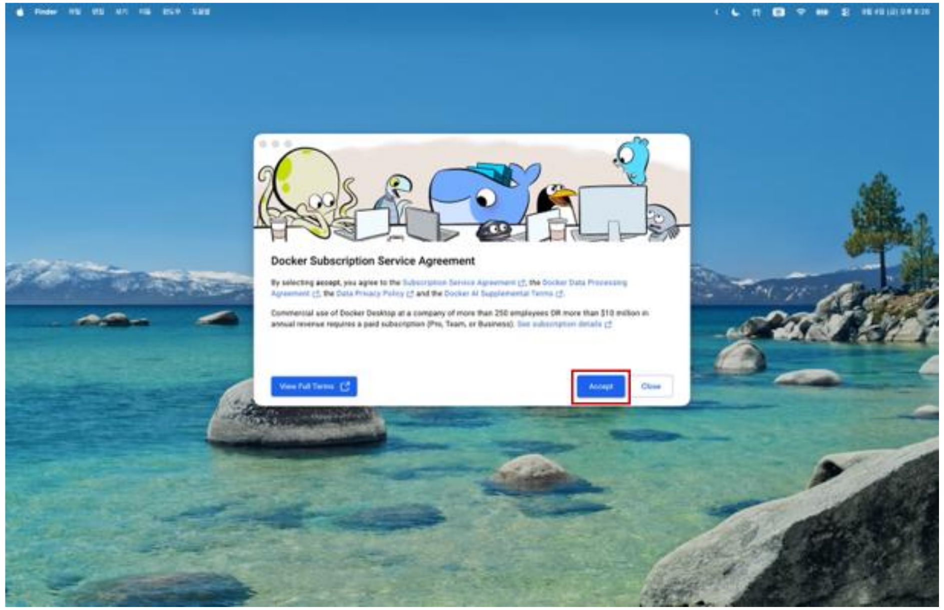


- If the Chip field shows an Apple M-series chip, your Mac uses **Apple silicon**.
- If the Processor field contains the word Intel, your Mac is an **Intel-based Mac**.

2. Install Docker Desktop

- Open the official Docker Desktop installation page for Mac.
  → https://docs.docker.com/desktop/setup/install/mac-install/
- Open Docker Desktop, accept the required terms, and use the recommended settings unless your institution instructs otherwise.

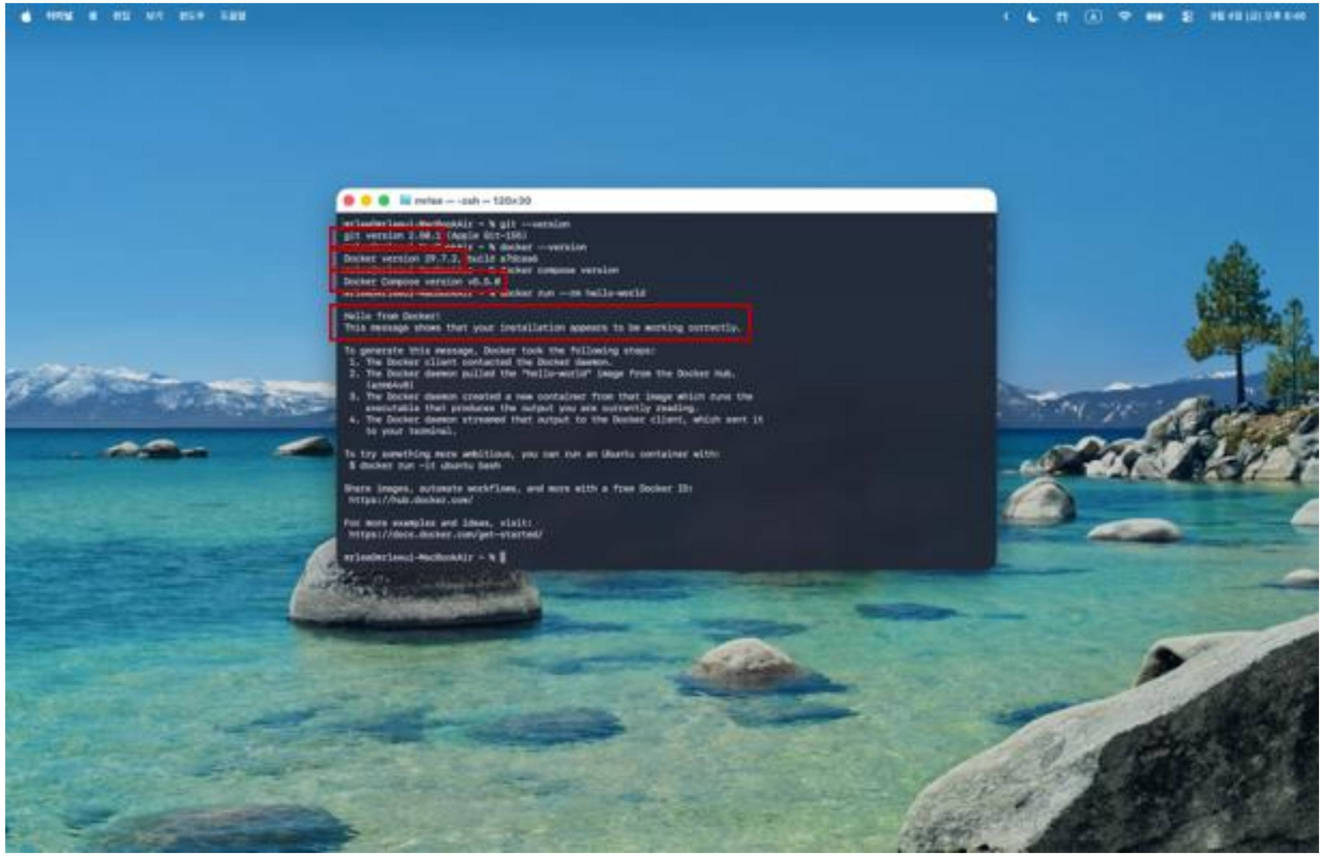




3. Verify the Required Tools

- Open Applications > Utilities > Terminal, then run:

```
git --version
docker --version
docker compose version
docker run --rm hello-world
```



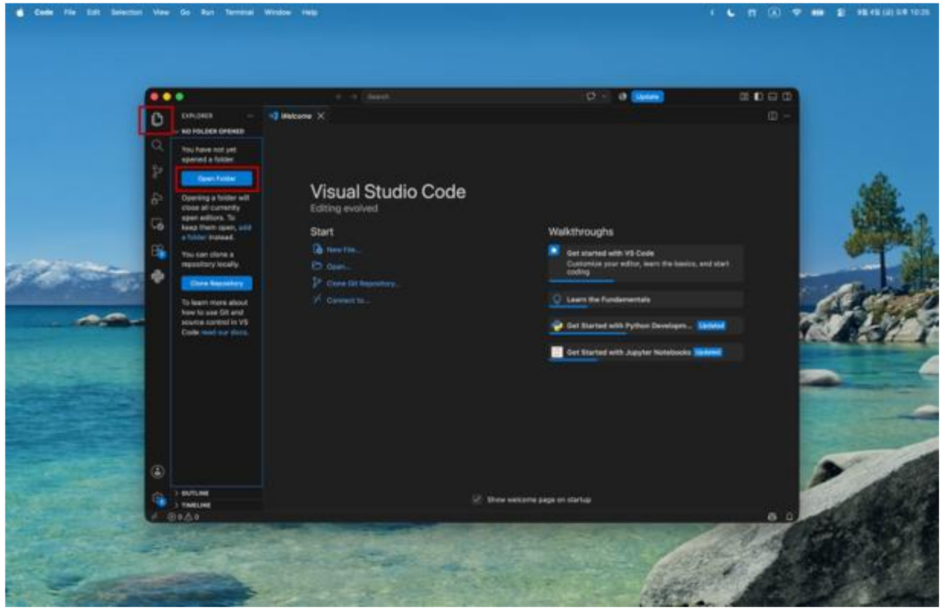
4. Clone the Course Repository

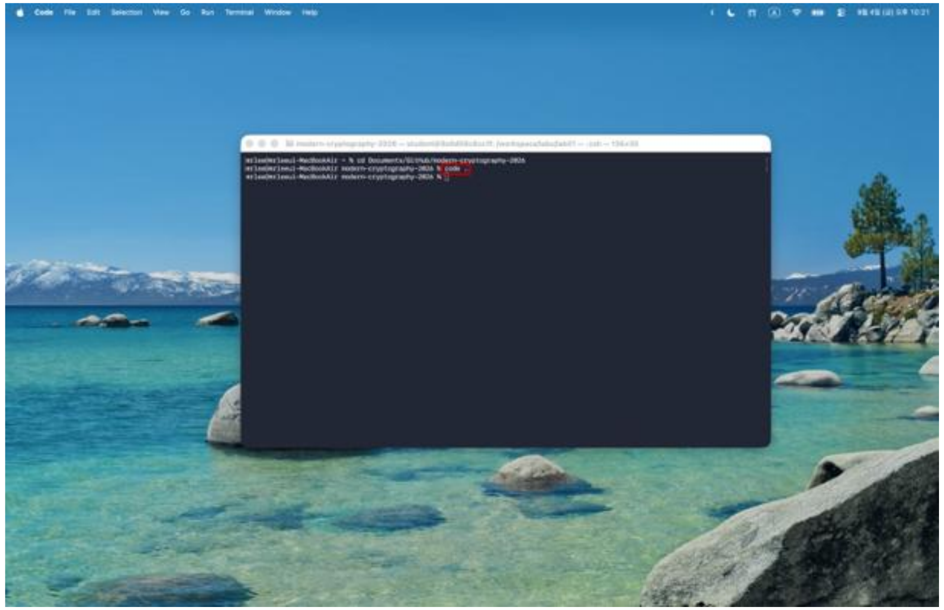
- **Run in: HOST - macOS Terminal**

```
git clone https://github.com/Wonsuk-EMSEC/modern-cryptography-2026.git
cd modern-cryptography-2026
```

5. Open the Course Repository in VS Code

- Open Visual Studio Code and open the cloned modern-cryptography-2026 folder.
- If the code command is available, navigate to the cloned repository directory, then run: `Code .`





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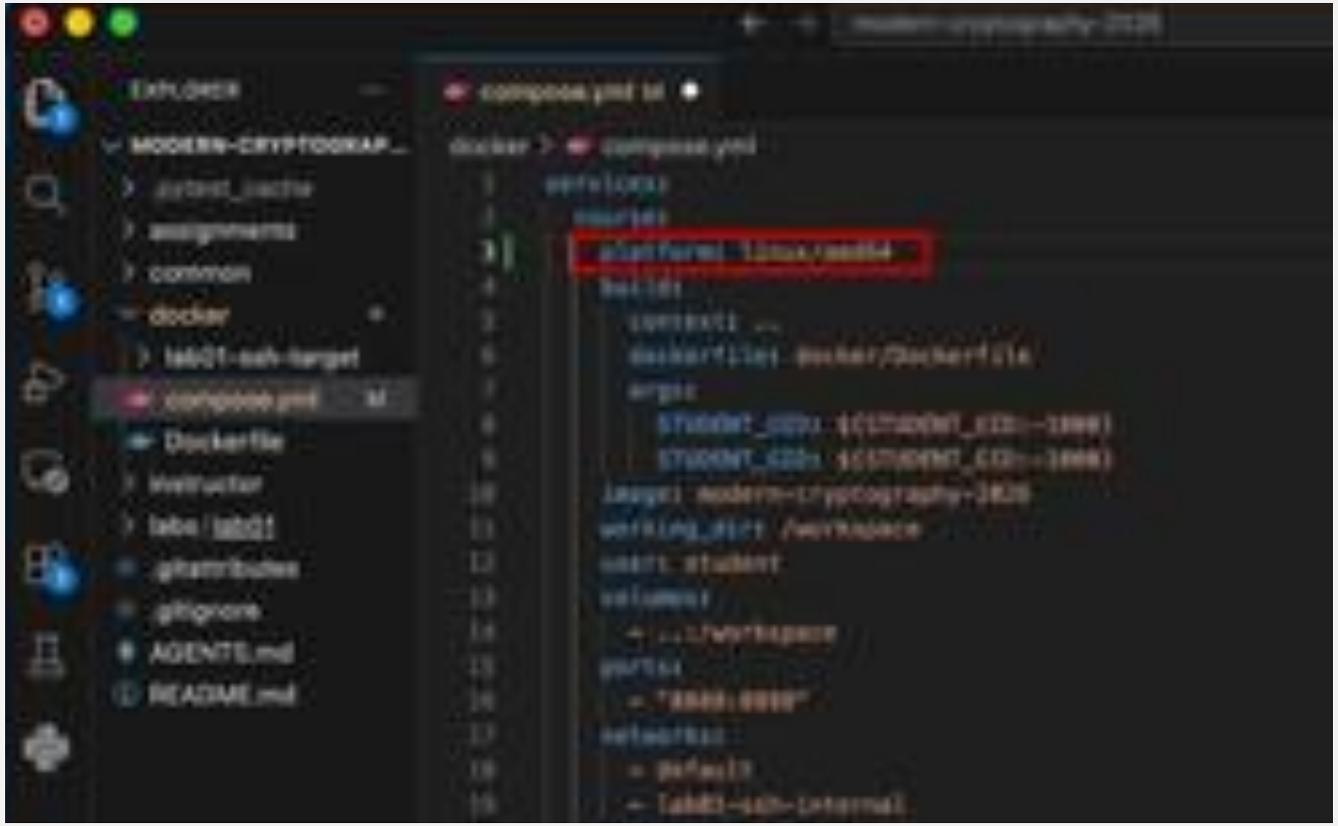
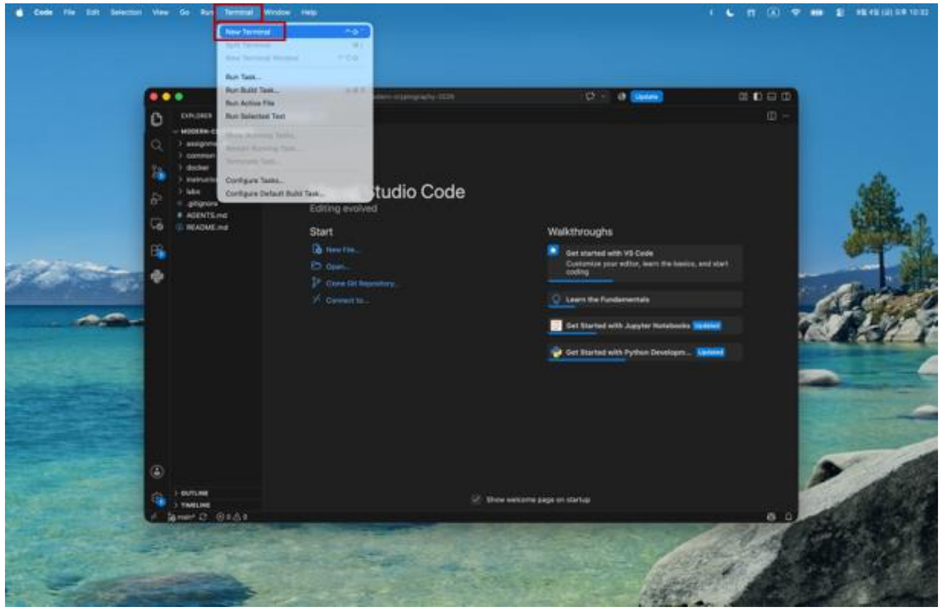
- Before building the course image, open docker/compose.yml and add:

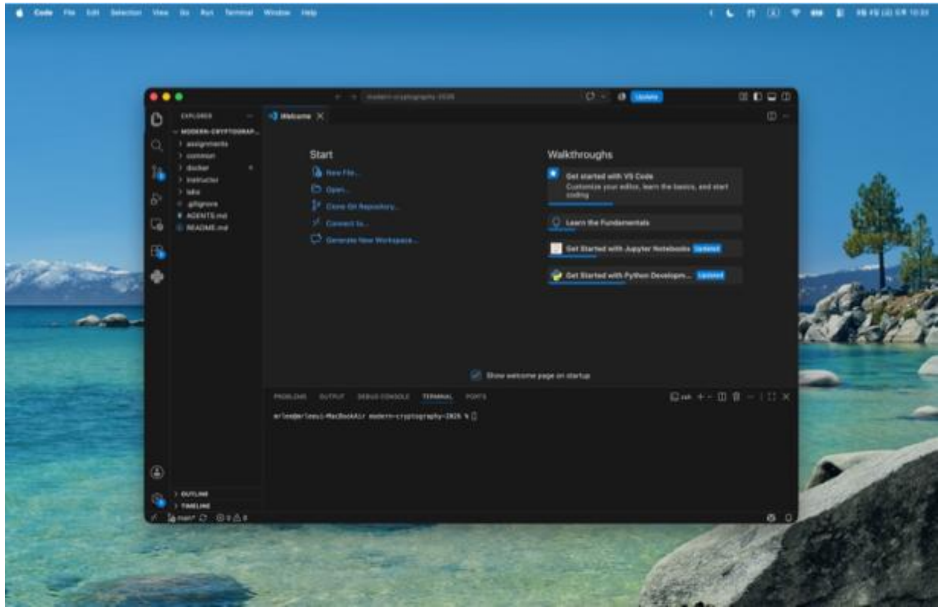
```
platform: linux/amd64
```



6. Open the Visual Studio Code Terminal

- In Visual Studio Code, select Terminal > New Terminal. Confirm that the terminal is in the repository root.





7. Build and Start the Course Container

- The first time you use the course environment, build the Docker image:

```
docker compose -f docker/compose.yml build
```

- After the image has been built, start the course container:

```
docker compose -f docker/compose.yml run --rm course bash
```

8. Test the Lab Environment

- Run:

```
cd /workspace/labs/lab01
pytest -q
```